

The background image shows a shallow, clear tide pool on a rocky shore. A large, orange-brown starfish is the central focus, partially covered by green and brown seaweed. The water is calm, reflecting the sky and the surrounding rocks. The overall scene is a natural, coastal environment.

Behavioral ecology and society

IB 431 – Behavioral Ecology
Fall 2025
December 2nd 2025

Outline

- I. History of eugenics and biology
- II. Biases in behavioral ecology
- III. Ethics of animal behavior research
- IV. Discuss Lyapun et al 2020 & Baptiste et al 2022

Eugenics

- “Movement” that seeks to “improve” the human species through selective breeding (late 1800s)
- Centers around inhibiting reproduction of “unfit” and promoting reproduction of more “fit” individuals
- Eugenic concepts have been repeatedly debunked



Eugenics and evolutionary biology

- Notable evolutionary biologists and geneticists pushed eugenics
 - Ronald Fisher
 - Francis Galton
 - JBS Haldane
- Universities also played a key role in promoting and funding eugenics



Eugenics in the United States

- Eugenics becomes mainstream in the U.S. by the 1910s
- Numerous states pass sterilization laws – targeting non-white groups and people with disabilities
- Anti-immigrant sentiment

54 *Popular Science Monthly*

Registering Human Pedigrees

How Kansas Develops Fitter Families;
A Remarkable Experiment in Eugenics

By Arthur Capper, U. S. Senator from Kansas



*Mrs. Mary T. Watts,
Audubon, Iowa*



*Dr. Florence B. Sherbon,
University of Kansas*



THE average American family of today is the result of haphazard mating. Men and women marry with little scientific thought as to their physical and mental fitness for bearing and rearing children. When the children come they too often are brought up in the same haphazard fashion in which their parents chose each other. Is it any wonder that the number of mentally and physically unfit increases?

For years, American farmers and breeders

member of a family. Another sheet of the record shows the results of the intelligence test, and an examination by a psychiatric specialist of the nerve reflexes, emotional and intellectual responses. A thorough structural examination is made, including strength tests and measurements. The medical record gives the results of a complete physical and organic examination, with blood pressure, hemoglobin blood test and the Wasserman blood test. Every member of the family is examined

The Eugenics Building at the Kansas Free Fair, where families are judged and registered. Seated in the center is a "fitter family" surrounded by examiners

Immigration Act of 1924

- Limited immigration from Eastern and Southern Europe and Asia
- Led by David Reed, a eugenics-supporting Senator
- Passed 62-6 in the Senate
- Eugencist Harry Laughlin calls bill a “breakthrough in the adoption of scientific racism as a foundation for immigration policy”
- Adolf Hitler also praises the bill



Eugenics and Nazi Germany

- Eugenics movement informed much of the ideology of Nazi Germany
- The Rockefeller Foundation funded German eugenics programs
 - Including the one that Josef Mengele worked in before he went to Auschwitz



Eugenics in the modern day

- Concepts from eugenics (e.g., racial superiority, population control) inform modern white nationalist ideology
- Often by twisting evolutionary and genetic concepts

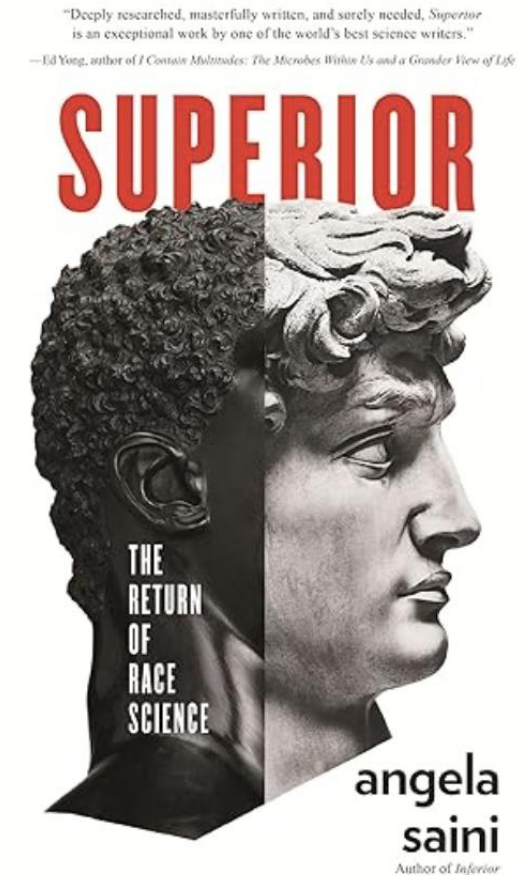


RESEARCH ARTICLE

How White nationalists mobilize genetics: From genetic ancestry and human biodiversity to counterscience and metapolitics

Aaron Panofsky✉, Kushan Dasgupta, Nicole Iturriaga

First published: 28 September 2020 | <https://doi.org/10.1002/ajpa.24150> | Citations: 66



Debunking eugenics (scientifically)

Debunking eugenics (scientifically)

- Eugenics represents a fundamental lack of understanding of the relationship between genes, behavior, and environment
 - Generally ignores environmental effects
 - Traits like IQ, mental health, behavior, etc. referenced in eugenics are driven by many loci each with small effect
 - Pleiotropy and genetic trade-offs also make most eugenics goals biologically impossible
 - Genetically human “races” are not distinct groups – most genetic variation in humans is within, not between populations

CRISPR and gene editing

- Given the historical context of eugenics, how do you feel about the potential of CRISPR and editing of genes in babies? What needs to be considered? In what contexts would you support (or not support) pursuing such practices? Talk with your tables for a few minutes.

NEWS RELEASE

World's First Patient Treated with Personalized CRISPR Gene Editing Therapy at Children's Hospital of Philadelphia

May 15, 2025



Biases in behavioral ecology research

- In this class or outside experience with the field, what biases have you noticed in behavioral ecology?

Biases in behavioral ecology research

- In this class or outside experience with the field, what biases have you noticed in behavioral ecology?
 - Geographic biases
 - Taxonomic biases
 - Identity/background of authors
 - Bias for positive / significant results
 - Perspective

White male perspectives dominate early naturalism and environmentalism movements

- Ability to travel freely was only available to rich and privileged individuals, typically just rich white men
- Focus on land use
- Ignoring or actively displacing indigenous communities



Same-sex sexual behavior in animals

- Historically viewed as an evolutionary conundrum – why waste energy on reproductive-type behavior
- Was this from a bias against such behaviors? Could there be alternative explanations?

[nature](#) > [nature ecology & evolution](#) > [perspectives](#) > article

Perspective | Published: 18 November 2019

An alternative hypothesis for the evolution of same-sex sexual behaviour in animals

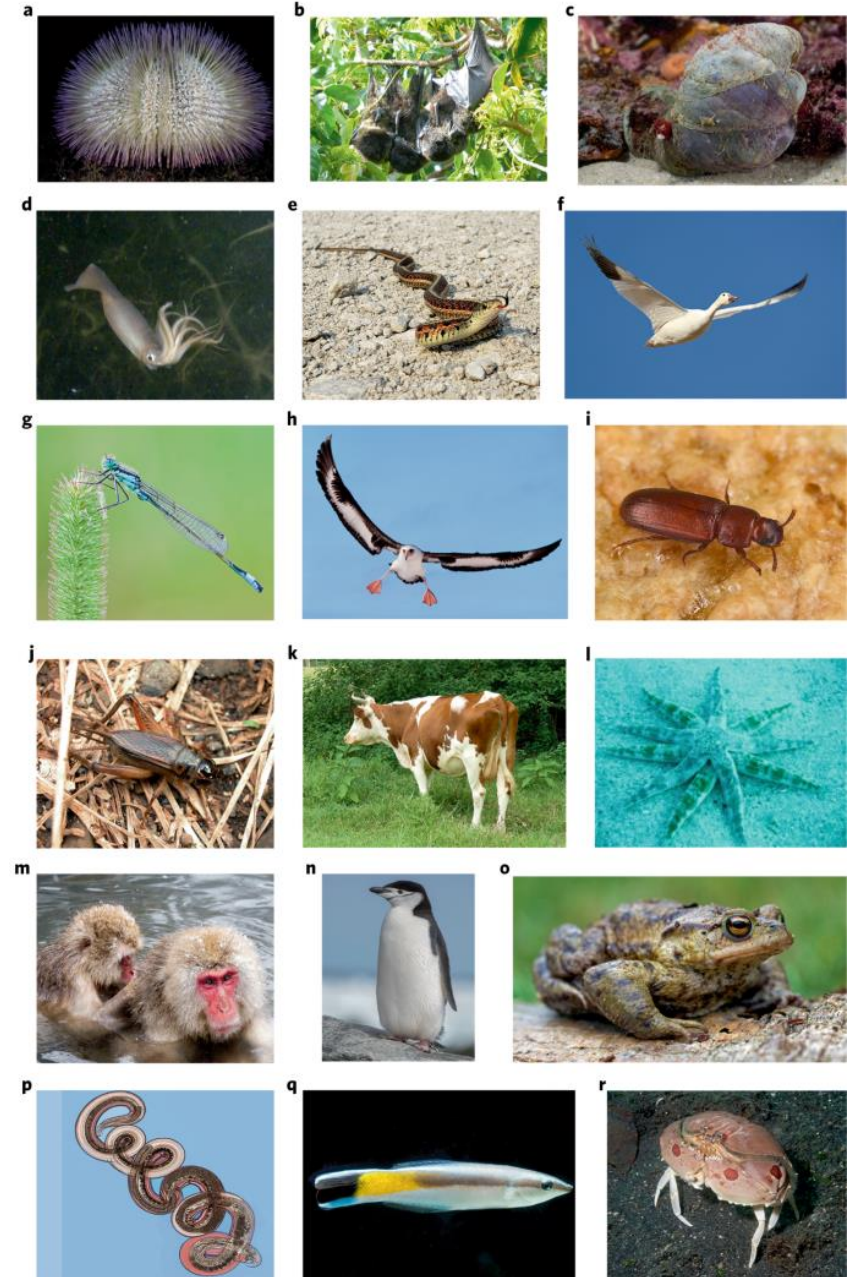
[Julia D. Monk](#) , [Erin Giglio](#), [Ambika Kamath](#), [Max R. Lambert](#) & [Caitlin E. McDonough](#)

[Nature Ecology & Evolution](#) **3**, 1622–1631 (2019) | [Cite this article](#)

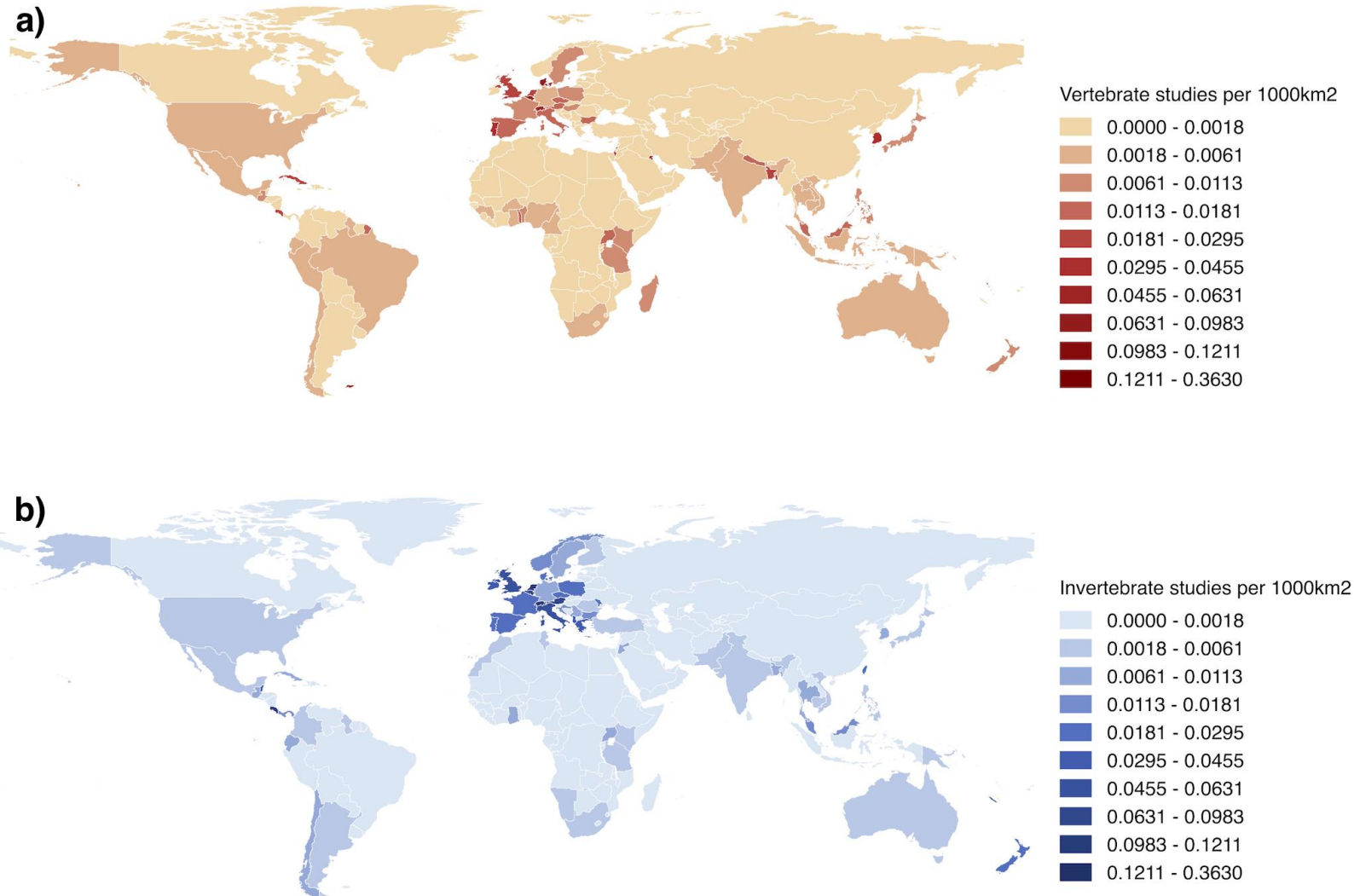
19k Accesses | **103** Citations | **1228** Altmetric | [Metrics](#)

Same-sex sexual behavior in animals

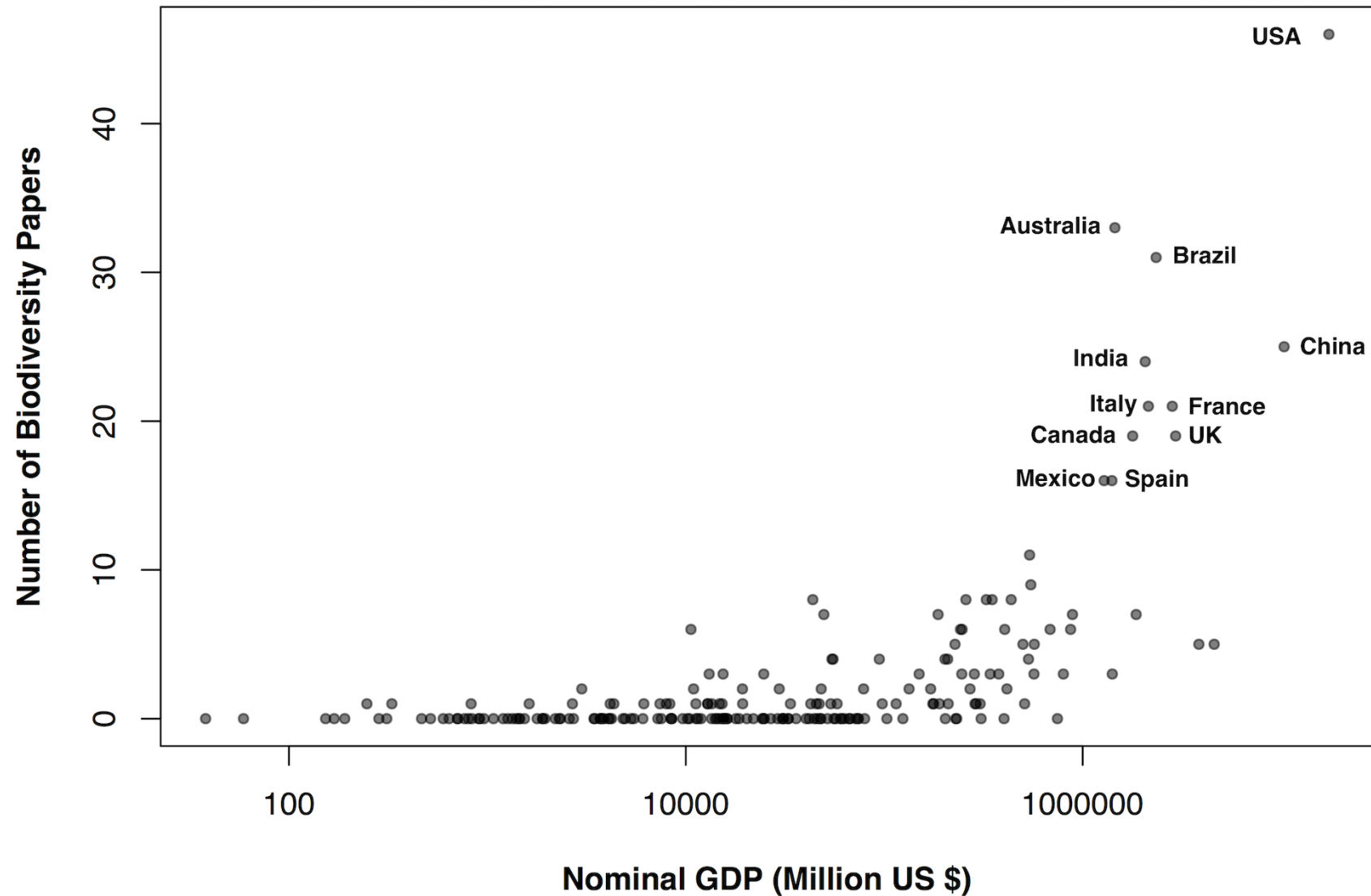
- Monk et al argue that indiscriminate sexual behavior is the ancestral state
- Suggest costs to SSB may be limited and exceeded by risks of missing reproductive event
- Highlight that evidence for SSB being costly is limited
- May be social benefits to SSB



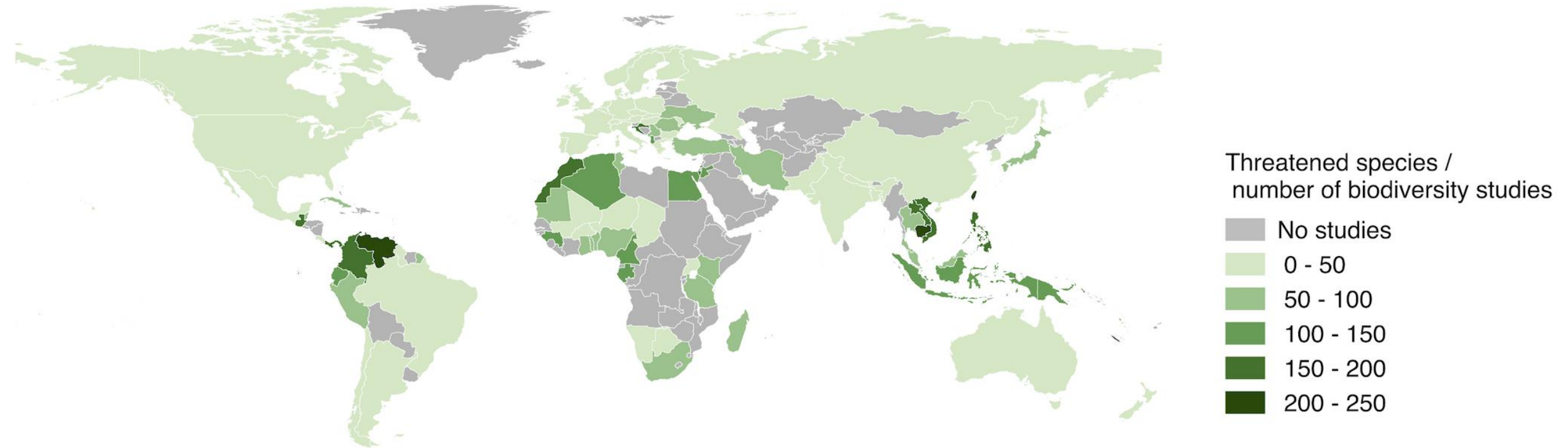
Bias for studies in NA and Europe



Bias for countries with higher GDP



Northern SA, Northern Africa, and Southeast Asia have few studies relative to # of threatened species



Reducing regional bias in research

- Take a few minutes to discuss with your tables.
 - How can we go about increasing data collection, collaboration, publication, etc. in these underrepresented regions without exploiting local communities?
 - In what contexts, if any, should “Western” scientists be able to take biological samples from these regions?
 - What role should indigenous communities play in biological research across the globe?

Citation bias

- Notable research can be underappreciated or under-credited
- Historically this often happened to authors from traditionally underrepresented groups in science

FRONTIERS IN ECOLOGY *and the* ENVIRONMENT

Research Communications |  Open Access |   

Historically excluded groups in ecology are undervalued and poorly treated

Richard B Primack, Tara K Miller, Carina Terry, Erika Marín-Spiotta, Pamela H Templer, Asmeret Asefaw Berhe, Emily J Diaz Vallejo, Meredith G Hastings, Vicki J Magley, Allison Mattheis, Blair B Schneider, Rebecca T Barnes✉ ... See fewer authors ^

First published: 07 March 2023 | <https://doi.org/10.1002/fee.2613> | Citations: 13

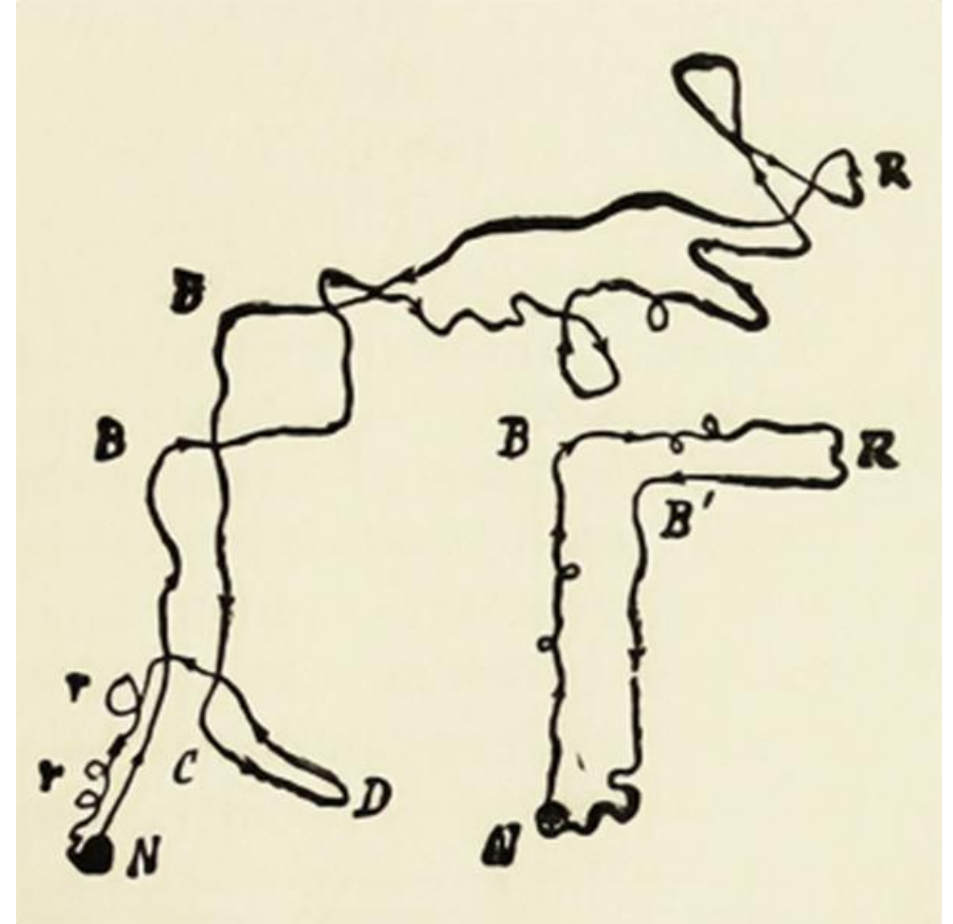
Charles Turner (1867 – 1923)

- Zoologist, entomologist, and psychologist
- Received PhD from the University of Chicago in 1923
- Published 70+ papers, focusing on social insects (bees, ants)
- All work conducted on his own while teaching high school
 - Could not get university job due to racist hiring practices



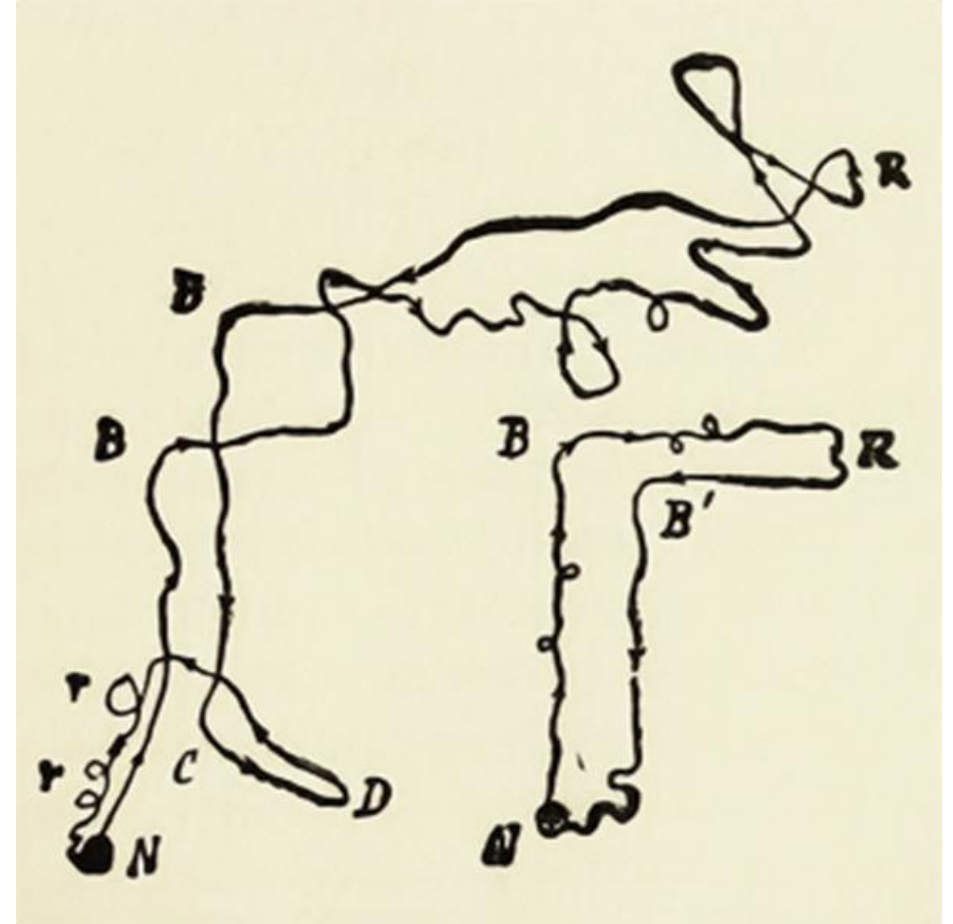
Charles Turner (1867 – 1923)

- First to consider cognitive capabilities of insects – conflicting with others of that era that thought animals acted solely on instinct
 - Spatial & visual learning in bees
 - Trial and error in cockroaches
- First to prove that insects are capable of hearing
- Among the first to explore collective behavior of insects



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Charles Turner (1867 – 1923)

- Needs to be recognized as one of the most influential figures in entomology and animal behavior
- Particularly in the study of cognition, social insects, and individual differences



Ethics in animal behavior research

- Pavlov's dog experiment was sketchy....electric shocks, surgically manipulating dog salivary glands
- Early macaque research isolated babies from their mothers



Ethics in animal behavior research

- Talk to your tables for a few minutes:
 - What other ethical concerns might arise in animal behavior research?
 - When (if ever) should lethal sampling be allowed in research?
 - When is it okay to keep animals in captivity? How invasive should field studies be allowed to be?



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